

Aryan Mishra

Berlin, Germany | +49 1745243900 | aryanmishraa12@gmail.com

Portfolio Website | GitHub | LinkedIn | My Blogs

Profile

Data science student with experience developing systems for automated data analysis, zero shot forecasting, and rag based document search. My work covers model evaluation, backend development, and deployment, alongside projects in robotics and reinforcement learning. I also enjoy building useful tools and sharing them as open-source software.

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, SQL, Bash.

AI & Data: PyTorch, JAX, Pandas, NumPy, Scikit-learn, LLM orchestration, RAG, FAISS, GraphRAG, LangChain, LlamaIndex, Docling, DeepAnalyze-8B, TimesFM, Chronos, Q-learning, Supabase.

Systems & Cloud: FastAPI, Docker, e2b, Ollama, llama.cpp, AWS, Supabase, GitHub Actions, n8n, Arduino, Raspberry Pi.

Professional Experience

CareerGenie

AI Engineer | Jan 2026 – Present

- Built a GitHub Signal Pipeline using Github REST API to turn users repository, commit, and pull-request activity into hiring signals, reducing overall tooling expenses by 10% through the elimination of a paid third-party API.
- Developed a career-benchmarking tool that evaluates applicant qualifications against target-role profiles, automatically quantifying skill gaps to deliver personalized progression roadmaps.

Teleperformance

Data Analytics Intern | Jun 2026 – Sep 2026

- Implemented the DeepAnalyze-8B framework into an autonomous agent that inspects datasets, executes end-to-end data science pipelines within an isolated sandbox, and synthesizes results into comprehensive reports.
- Implemented a zero-shot forecasting pipeline for workforce management using TimesFM, reducing sMAPE by **32%** compared to the internal ensemble baseline without requiring retraining.

Fertrado

AI Engineer Intern | Sep 2025 – Nov 2025

- Built an internal assistant for EU fertilizer regulations using Qwen, Ollama, and FastAPI; deployed the containerized application on AWS.
- Parsed regulatory documents and indexed them with FAISS to ground answers in relevant sources and citations; designed ingestion to support additional policy frameworks.

Infovirgin Technology Solutions Pvt. Ltd.

Data Science & AI Intern | Mar 2025 – May 2025

- Automated data preparation with a Python ETL pipeline and KNN imputation, cutting weekly preparation time by 75% and eliminating manual data-entry discrepancies.
- Built real-time dashboards for sales, client acquisition, and operations; standardized KPI definitions and reorganized analytics tables into a star schema covering 45+ attributes.
- Developed ARIMA and rolling-average models for revenue and workload forecasting, implementing randomized rolling-window search to cut training runtimes from hours to minutes and improve engineering capacity planning.

Projects

A.D.S.A. — Autonomous Data Science Agent

Python, DeepAnalyze-8B, Docker, e2b, Ollama

github.com/aryanwise/adsa

- Implemented an autonomous data science agent with the help of DeepAnalyze-8B research paper that automates the full workflow: data exploration, cleaning, feature engineering, and model training.
- Secured dynamic code execution inside an e2b/Docker sandbox, adding automated self-debugging to fix runtime errors and block unsafe system operations.
- Automated the delivery of complete technical summaries, compiling chart visualizations, model performance scores, and key analytical takeaways.

OpenbookLM — Local-First Research and Code Assistant

TypeScript, FastAPI, C++, GraphRAG, llama.cpp, USearch

github.com/aryanwise/OpenbookLM

- Working on a fully local research workspace to query technical papers, documentation, and source code with zero external data leakage.
- Implementing a graph-based retrieval (GraphRAG) and syntax-aware indexing to navigate complex code-bases and automate multi-file refactoring.
- Integrating local LLM inference using llama.cpp and vectorized search to generate, test, and verify code modifications completely offline.

Polymarket Analyzer — Prediction Market Intelligence

Python, Polymarket API, Supabase, Groq, Gemini

github.com/aryanwise/Polymarket-Analysis-BIT-Capital

- Built an automated ETL pipeline using GitHub Actions to ingest prediction market odds, financial news, and equity metrics at dynamic intervals (5m–5h).
- Integrated Groq and Gemini LLMs to synthesize real-time market sentiment, probability shifts, and price targets into actionable executive briefs.
- Compared prediction market odds with Yahoo Finance stock data in Supabase, using a 6-hour cache to evaluate prediction accuracy.

Autonomous 4WD Obstacle-Avoidance Rover

C++, Arduino, L298N Driver, Ultrasonic Sensor

github.com/aryanwise/autonomous-obstacle-avoidance-robot

- Built an autonomous 4WD rover executing real-time spatial navigation and obstacle avoidance via multi-angle ultrasonic scanning and state-machine decision logic in C++.
- Implemented dual H-bridge motor control with PWM speed regulation, optimizing power delivery and eliminating sensor-polling latency during active maneuvers.

Zombie Shooter RL — Reinforcement Learning

Python, Pygame, NumPy, Q-Learning

github.com/aryanwise/Reinforcement-learning-on-Game

- Built a custom 2D simulation environment and trained a tabular Q-learning agent over 5,000+ episodes, engineering state discretization for continuous spatial coordinates and enemy proximity vectors.
- Formulated a shaped reward architecture balancing evasion and targeting, evaluating policy convergence and action distributions via reward curves and Q-value heatmaps.

Gomoku Game AI — Heuristic Search Engine

Python, Minimax, Alpha-Beta Pruning, Iterative Deepening

github.com/aryanwise/Gomoku-AI-Game

- Built a real-time Gomoku (game AI agent) operating under 10-second turn constraints, combining iterative deepening minimax with alpha-beta pruning to dynamically maximize search depth.
- Implemented multi-directional pattern threat evaluation and local-neighborhood move pruning, drastically reducing tree branching factor while prioritizing winning sequences.

Education

GISMA University of Applied Sciences | Berlin, Germany

B.Sc. in Data Science, AI & Digital Business | Oct 2023 – Apr 2027

Interests

Swimming, Reading, and Touching Grass.

Languages

English - Fluent | German - A2 | Hindi - Native Speaker